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# GCSE Computer Science – Programming Data Structures (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a data structure?

- ○ **A.** A way of organising and storing data so it can be accessed and modified efficiently
- ○ **B.** A physical component inside a computer that stores data
- ○ **C.** A type of programming language
- ○ **D.** A diagram showing how a program works

Q2 (1 mark)

What is an array?

- ○ **A.** A data structure that stores multiple values of the same data type under one identifier
- ○ **B.** A single variable that can only store one value at a time
- ○ **C.** A type of loop used to repeat code
- ○ **D.** A function that returns multiple values

Q3 (1 mark)

In most programming languages, what is the index of the first element in an array?

- ○ **A.** 1
- ○ **B.** 0
- ○ **C.** -1
- ○ **D.** The length of the array

Q4 (1 mark)

What is a record in programming?

- ○ **A.** A data structure that groups together related items of potentially different data types under a single identifier
- ○ **B.** A log file that stores error messages
- ○ **C.** A type of loop that records how many times code runs
- ○ **D.** A backup copy of a program

Q5 (1 mark)

What is a field in the context of records?

- ○ **A.** The memory address where a record is stored
- ○ **B.** The total number of records in a database
- ○ **C.** A type of error that occurs in records
- ○ **D.** A named item within a record that stores a single piece of data

Q6 (1 mark)

What is a two-dimensional array?

- ○ **A.** An array that can only store two elements
- ○ **B.** An array that stores data in a grid format with rows and columns
- ○ **C.** Two separate arrays that are linked together
- ○ **D.** An array that uses twice as much memory

Q7 (1 mark)

An array called `temperatures` contains 7 values. Using zero-based indexing, what is the index of the last element?

- ○ **A.** 7
- ○ **B.** 8
- ○ **C.** 6

- **D.** 1

Q8 (1 mark)

How do you typically access a field called `name` from a record variable called `student`?

- **A.** name.student
- **B.** student[name]
- **C.** student.name
- **D.** name(student)

Q9 (1 mark)

What is one advantage of using an array instead of multiple separate variables?

- **A.** Arrays allow you to process multiple related values using loops, making code more efficient and maintainable
- **B.** Arrays can only store one value, making code simpler
- **C.** Arrays run faster than any other data structure
- **D.** Arrays do not require any memory to store data

Q10 (1 mark)

Which scenario would be best suited to using a record data structure?

- ○ **A.** Storing a list of 50 temperature readings, all as real numbers
- ○ **B.** Storing a single student's test score as an integer
- ○ **C.** Counting from 1 to 100
- ○ **D.** Storing information about a single product: name (String), price (Real), stockLevel (Integer), isAvailable (Boolean)

Q11 (1 mark)

In a two-dimensional array, how many index values are needed to access a single element?

- ○ **A.** One
- ○ **B.** Two
- ○ **C.** Three
- ○ **D.** None - elements are accessed by name

Q12 (1 mark)

Which of the following correctly declares a one-dimensional array called `ages` containing 5 integer values?

- ○ **A.** ages <- [25, 30, 22, 28, 35]
- ○ **B.** ages <- 25, 30, 22, 28, 35
- ○ **C.** INTEGER ages <- 5
- ○ **D.** ages = ARRAY(String, 5)

Q13 (2 marks)

Given the array `colours <- ["Red", "Green", "Blue", "Yellow", "Purple"]`, what is output by the following code?

```
OUTPUT colours[1]
OUTPUT colours[4]
```

- ○ **A.** Green and Purple
- ○ **B.** Red and Purple
- ○ **C.** Red and Yellow
- ○ **D.** Green and Yellow

Q14 (2 marks)

What is output by the following code?

```
numbers <- [10, 20, 30, 40]
FOR i <- 0 TO 3
    OUTPUT numbers[i]
ENDFOR
```

- ○ **A.** 10 only
- ○ **B.** 40 only
- ○ **C.** 0, 1, 2, 3
- ○ **D.** 10, 20, 30, 40

Q15 (2 marks)

A school wants to store information about each teacher: their name, subject taught, and years of experience. Which record definition is correct?

- ○ **A.** ARRAY Teacher[name, subject, years]
- ○ **B.** RECORD Teacher name: String subject: String yearsExperience: Integer ENDRECORD
- ○ **C.** Teacher <- [name, subject, years]
- ○ **D.** FUNCTION Teacher(name, subject, years)

Q16 (2 marks)

Given this record and instance, what is output?

```
RECORD Book
    title: String
    author: String
    pages: Integer
ENDRECORD

myBook <- Book("1984", "George Orwell", 328)
OUTPUT myBook.author
```

- ○ **A.** 1984
- ○ **B.** Book
- ○ **C.** 328
- ○ **D.** George Orwell

Q17 (2 marks)

A 2D array represents a seating plan:

```
seats <- [["A1", "A2", "A3"],
          ["B1", "B2", "B3"],
          ["C1", "C2", "C3"]]
```

What is the value of `seats[1][2]`?

- **A.** A1
- **B.** A3
- **C.** B3
- **D.** C2

Q18 (2 marks)

What are the contents of the array after this code executes?

```
data <- [5, 10, 15, 20]
data[2] <- 100
```

- **A.** [5, 10, 15, 20]
- **B.** [100, 10, 15, 20]
- **C.** [5, 100, 15, 20]
- **D.** [5, 10, 100, 20]

Q19 (2 marks)

What is output by this code?

```
values <- [3, 7, 2, 8]
total <- 0
```

```
FOR i <- 0 TO 3
    total <- total + values[i]
ENDFOR
OUTPUT total
```

- A. 3
- B. 8
- C. 20
- D. 4

Q20 (2 marks)

What is output after this code executes?

```
RECORD Student
    name: String
    score: Integer
ENDRECORD

pupil <- Student("Amy", 75)
pupil.score <- 82
OUTPUT pupil.score
```

- A. 75
- B. Student
- C. Amy
- D. 82

Q21 (2 marks)

To process every element in a two-dimensional array, what type of loop structure is typically used?

- ○ **A.** A single FOR loop
- ○ **B.** A single WHILE loop
- ○ **C.** Nested loops (one loop inside another)
- ○ **D.** No loop is needed

Q22 (2 marks)

This code searches an array for a target value. What is output when `target <- 15`?

```
numbers <- [10, 25, 15, 30, 5]
found <- False
FOR i <- 0 TO 4
    IF numbers[i] = target THEN
        found <- True
    ENDIF
ENDFOR
OUTPUT found
```

- ○ **A.** True
- ○ **B.** False
- ○ **C.** 15
- ○ **D.** 2

Q23 (2 marks)

Why might a programmer use an array of records?

- ○ **A.** To store a single item with multiple attributes
- ○ **B.** To store multiple items where each item has multiple attributes of different types
- ○ **C.** To make the program run faster
- ○ **D.** Because arrays cannot store any other data type

Q24 (3 marks)

What is output by this code?

```
grid <- [[1, 2, 3],
         [4, 5, 6]]
total <- 0
FOR row <- 0 TO 1
    FOR col <- 0 TO 2
        total <- total + grid[row][col]
    ENDFOR
ENDFOR
OUTPUT total
```

- ○ **A.** 6
- ○ **B.** 15
- ○ **C.** 21
- ○ **D.** 12

Q25 (3 marks)

This code finds the position of the largest value in an array. What is output?

```
values <- [45, 72, 38, 91, 63]
maxIndex <- 0
FOR i <- 1 TO 4
    IF values[i] > values[maxIndex] THEN
        maxIndex <- i
    ENDIF
ENDFOR
OUTPUT maxIndex
```

- ○ **A.** 91
- ○ **B.** 0
- ○ **C.** 4
- ○ **D.** 3

Q26 (3 marks)

Given this code, what is output?

```
RECORD Product
    name: String
    price: Real
ENDRECORD

items <- [Product("Pen", 1.50), Product("Book", 8.99), Product("F
total <- 0
FOR i <- 0 TO 2
    total <- total + items[i].price
ENDFOR
OUTPUT total
```

- ○ **A.** 3
- ○ **B.** 11.24
- ○ **C.** Pen, Book, Ruler

- D. 8.99

Q27 (3 marks)

A 2D array stores a game board where 1 means occupied and 0 means empty:

```
board <- [[1, 0, 1],
          [0, 1, 0],
          [1, 1, 0]]
```

How many occupied spaces (1s) are in the board?

- A. 4
- B. 5
- C. 6
- D. 9

Q28 (3 marks)

What are the contents of the array after this code executes?

```
letters <- ["A", "B", "C", "D"]
temp <- letters[0]
letters[0] <- letters[3]
letters[3] <- temp
```

- A. ["A", "B", "C", "D"]
- B. ["A", "B", "C", "A"]
- C. ["D", "B", "C", "D"]

- **D.** ["D", "B", "C", "A"]

Q29 (3 marks)

A program stores user accounts as records. Analyse which record definition is most appropriate for storing: username, email address, year of birth, and whether the account is active.

- **A.** RECORD User username: Integer email: Integer yearOfBirth: String isActive: String ENDRECORD
- **B.** RECORD User username: String email: String yearOfBirth: Integer isActive: Boolean ENDRECORD
- **C.** ARRAY User[username, email, yearOfBirth, isActive]
- **D.** RECORD User data: String ENDRECORD

Q30 (3 marks)

A school stores student marks in a 2D array where rows represent students (4 students) and columns represent subjects (3 subjects). Analyse what `marks[2][1]` represents.

- **A.** The mark for the first student in the second subject
- **B.** The mark for the second student in the first subject
- **C.** The mark for the third student in the second subject
- **D.** The total marks for student 2

Total: **55** marks • Time: **30 min**

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